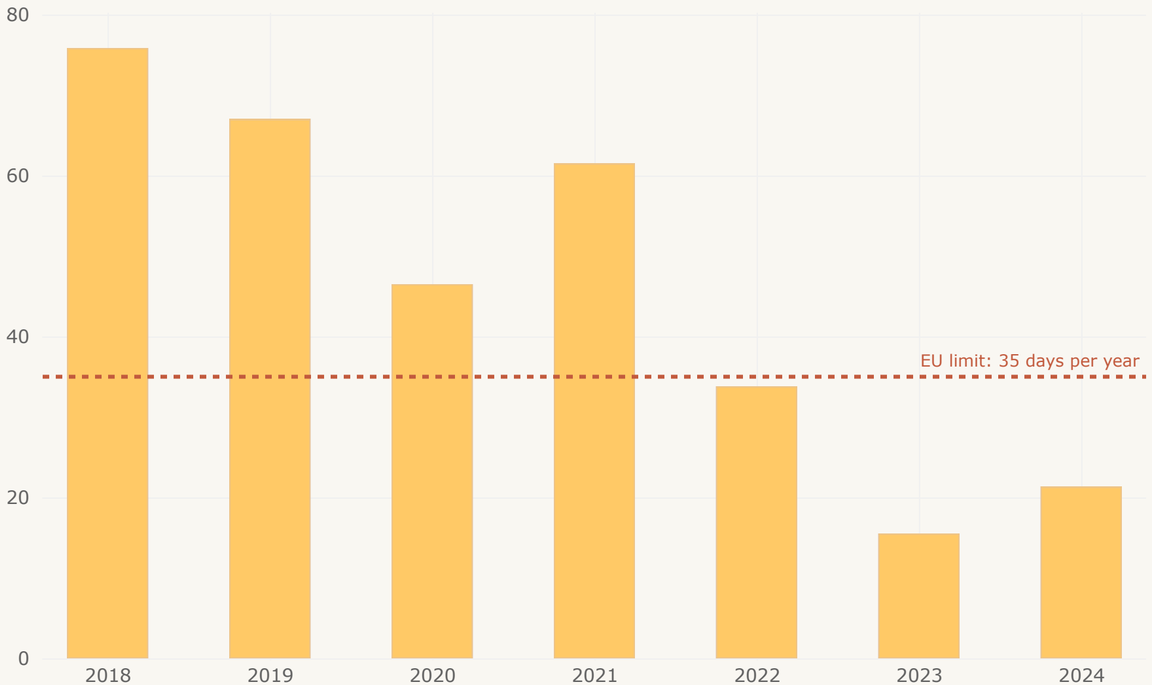
# Air quality: Kraków

City: Kraków

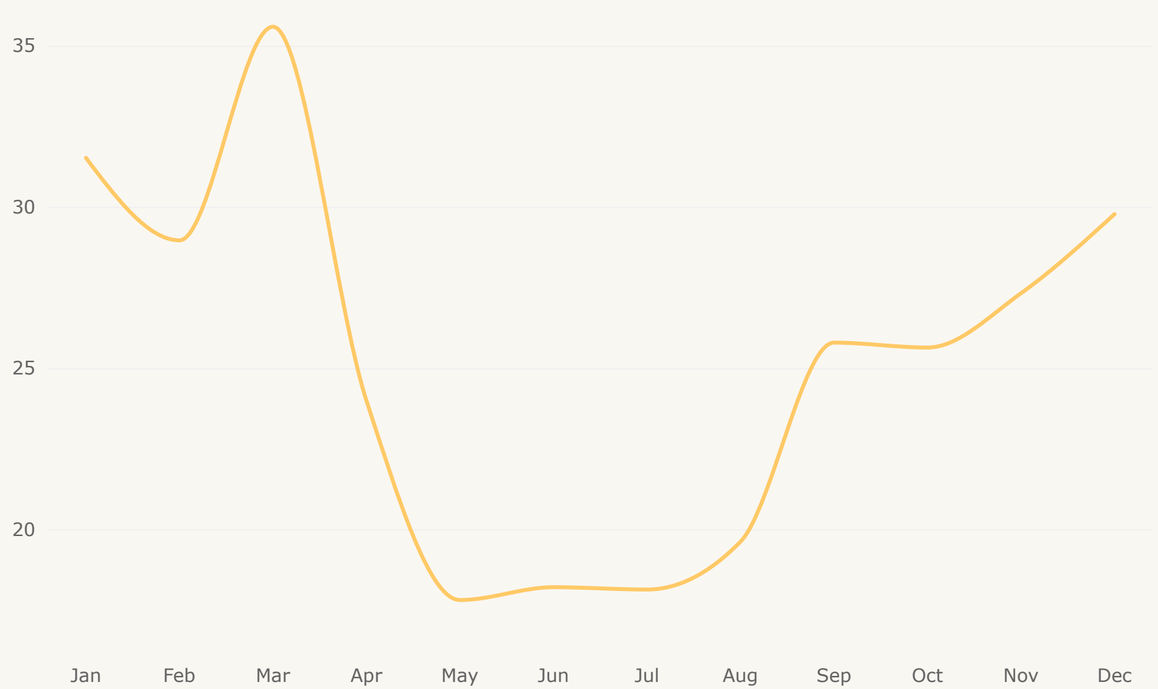
Year: 2024

| Average PM10 (µg/m³) | Days above PM10 limit | Change in smog days vs 2018 | Benzo(a)pyrene (ng/m³, limit: 1) |
|---|---|---|---|
| 25.2 | 21.3 | -71.87% | 1.11 |

## Days above the PM10 daily limit (50 µg/m³) by year




## Average PM10 by month

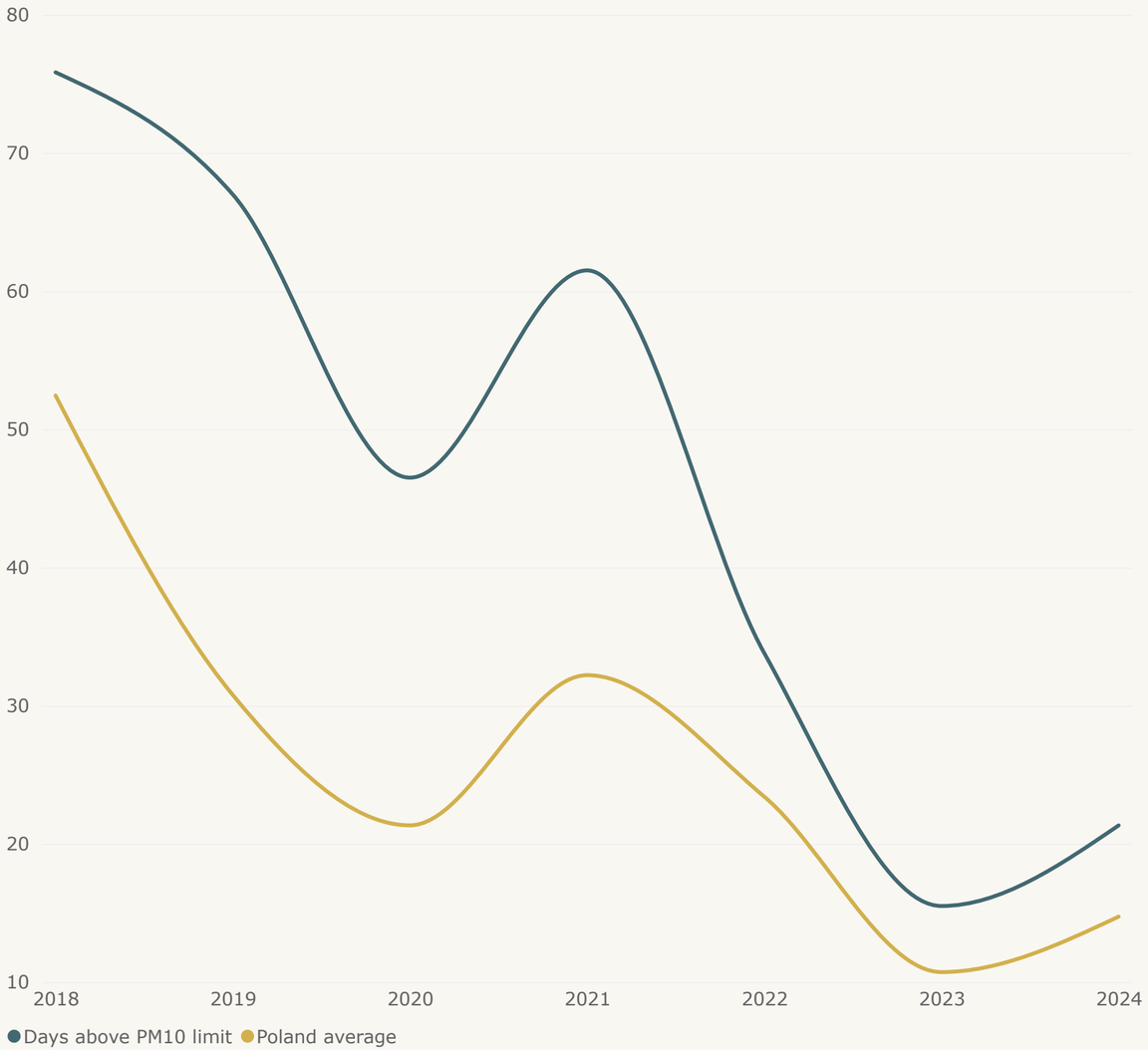


# Air quality: Kraków

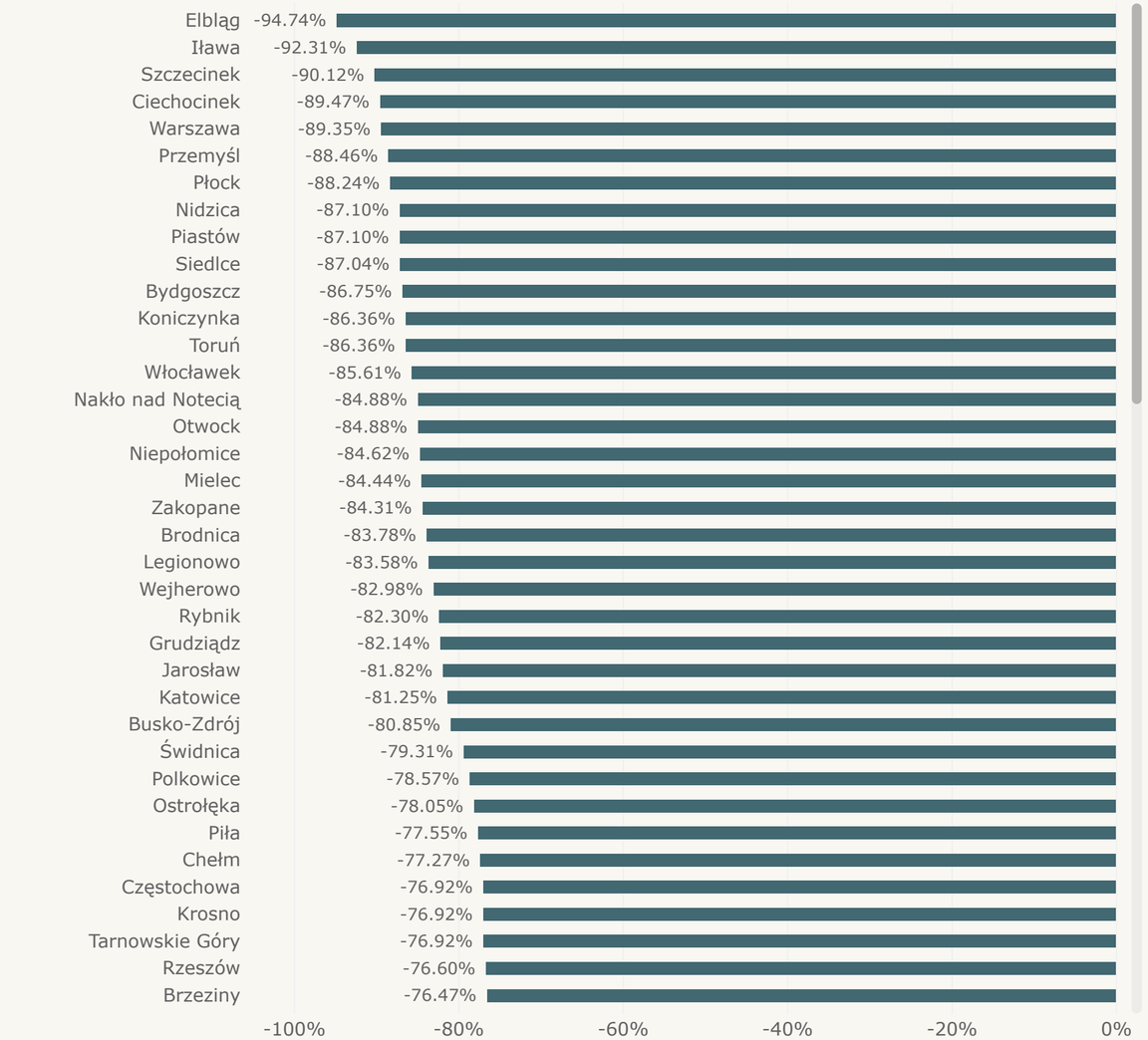
City: Kraków

Year: 2024

## Days above the PM10 limit: selected city vs Poland average




## Change in smog days vs 2018 (cities above the 35-day limit in 2018)



# Air quality: Kraków

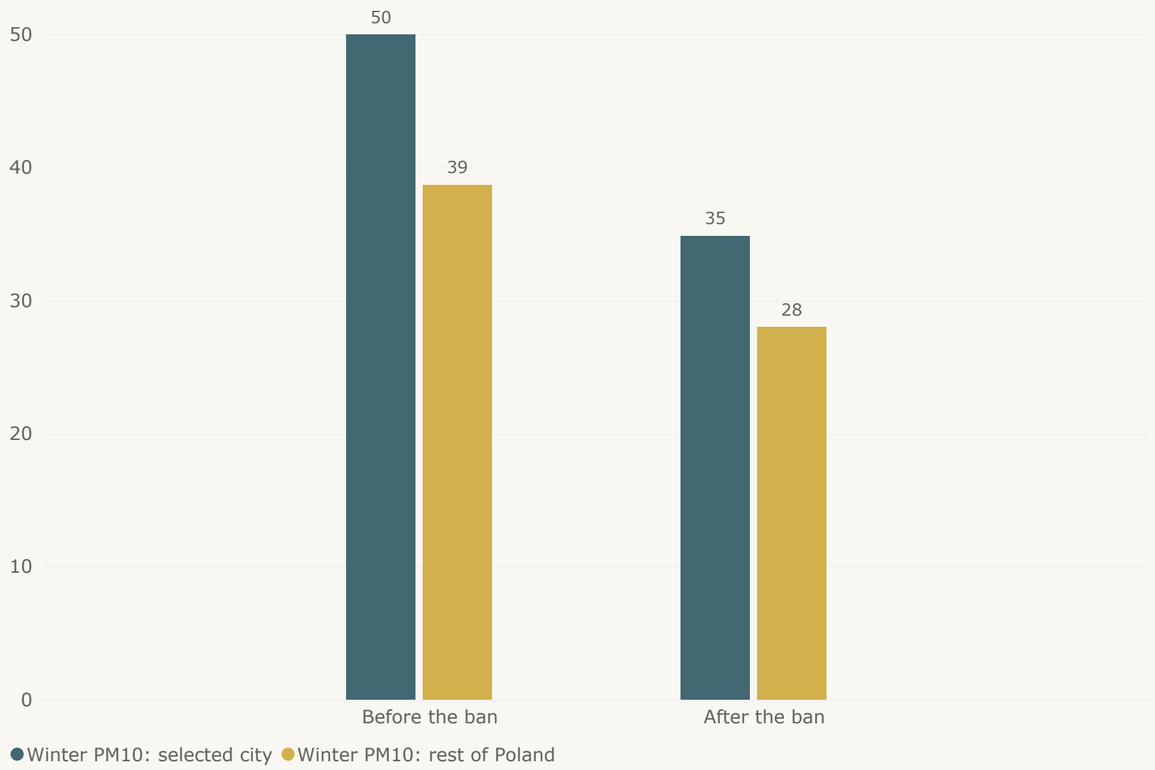
City: Kraków

Winter PM10 change: selected city
-30.29%

Winter PM10 change: rest of Poland
-27.67%

Ban effect (percentage points)
-2.6 p.p.

Average PM10 in the heating season (Oct–Mar): before and after the ban




**Question**
Did the ban on burning coal in Kraków (in force since September 2019) improve air quality more than in the rest of Poland?

**Method**
Average PM10 in the heating season (Oct–Mar) before and after the ban, in Kraków and in all other cities (a simplified difference-in-differences design).

**Result (Kraków)**
Kraków: -30%, rest of Poland: -28%. The difference is only about 2.6 percentage points, so the 2018–2024 data shows no large additional effect of the ban itself. Kraków had been replacing coal stoves for years before 2019, so much of the improvement may have happened before this data starts. The similar drop across Poland also points to common factors: weather and nationwide programmes.

**Limitations**
Only about 1.5 heating seasons before the ban, a changing set of stations, no weather data.

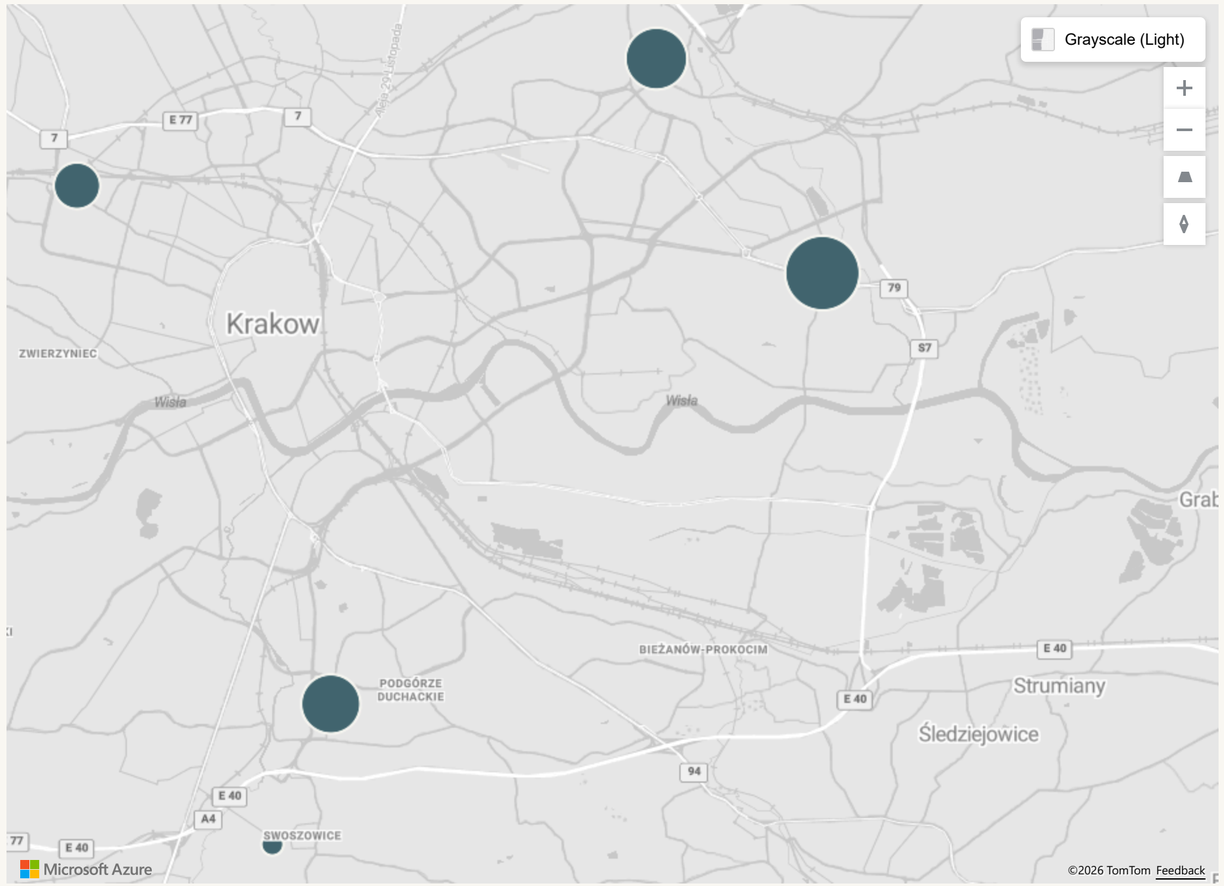
**Recommendations**
- Evaluate policies against a control group, not only a base year. The 72% drop in smog days looked like a big success of the ban, but most of that change happened elsewhere too.
- Kraków still has ~25% higher winter PM10 than the rest of Poland. Further action could target transport and industry (worst station in 2024: Nowa Huta, ul. Bulwarowa).
- Benzo(a)pyrene is still above the limit (1.11 ng/m³). Pollution drifting in from municipalities around Kraków, where the ban does not apply, is worth investigating.

# Air quality: Kraków

City: Kraków

Year: 2024

## Monitoring stations: days above the PM10 limit in the selected year




| Station | Days above PM10 limit | Average PM10 (µg/m³) |
|---|---|---|
| Kraków, ul. Bulwarowa | 30.0 | 28.5 |
| Kraków, os. Piastów | 24.0 | 26.4 |
| Kraków, ul. Bujaka | 23.0 | 25.2 |
| Kraków, ul. Złoty Róg | 19.0 | 25.1 |
| Kraków, os. Swoszowice | 16.0 | 22.1 |
| Kraków, os. Wadów | 16.0 | 23.8 |
| **Total** | **21.3** | **25.2** |

# Data quality: what the app detected and fixed

Number of measurements
960K

Number of stations
296

| Year | BaP(PM10) | PM10 | PM2.5 | **Total** |
|---|---|---|---|---|
| 2018 | 45928 | 58852 | 23319 | **128099** |
| 2019 | 53925 | 63521 | 22734 | **140180** |
| 2020 | 54826 | 61254 | 23527 | **139607** |
| 2021 | 57378 | 62693 | 20728 | **140799** |
| 2022 | 58419 | 62308 | 19972 | **140699** |
| 2023 | 53682 | 61178 | 19116 | **133976** |
| 2024 | 60582 | 56584 | 19166 | **136332** |
| **Total** | **384740** | **426390** | **148562** | **959692** |

**Fixed by the app (Power Query Cleaner)**
• Wide layout: stations in columns and a 3-row header → unpivoted into a long table (date, station, value)
• Old station codes: GIOŚ renamed stations over the years → old codes automatically mapped to current ones using the metadata
• Numbers stored as text with a decimal comma (2018 BaP file) → converted with Polish regional settings

**Found while building the model**
• Duplicate codes in the station metadata → they blocked the model relationship, removed
• -999 used instead of missing coordinates → replaced with blank
• Stations with no measurements in a given year → excluded from comparisons (they caused a false "-100%")

**Next step:** add detection of duplicate keys and missing-value codes (e.g. -999) to the app.